Ryan Jan

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Education

California Polytechnic State University — San Luis Obispo, CA

Bachelor of Science in Electrical Engineering, Minor in Mathematics | GPA: 3.67/4.00 — *Expected June 2029*

- **Relevant Coursework:** Electric Circuit Analysis I–II, Signals and Systems, Introduction to Computer Systems, Introduction to HDL & Digital Design, Nonlinear Dynamical Systems, Complex Analysis

Engineering Experience

Cal Poly CubeSat Laboratory — San Luis Obispo, CA

Avionics Member — *Oct. 2026 – Present*

- Contributed to bring-up and hardware/software checkout of the mZ Rev. 10 CubeSat avionics PCB through visual and continuity inspection, controlled first power-on, and power-rail verification.
- Programmed and verified I2C FRAMs using embedded Linux tools and validated multiplexed buses, onboard sensors, and board-level interfaces during spacecraft hardware checkout.
- Developing a proprietary avionics debug PCB for internal hardware testing and bring-up, contributing to schematic-level electrical design and component/interface decisions.

PROVE Lab – Cal Poly Prototype Vehicles Laboratory — San Luis Obispo, CA

Autonomy Team Member — *Fall 2026 – Present*

- Built an NVIDIA Isaac Sim and ROS 2 sensor-simulation pipeline for an autonomous vehicle, publishing simulated IMU and magnetometer measurements for localization development.
- Developed Python ROS 2 tooling for sensor noise and bias injection and automated PhysX vehicle tests; debugged sensor frames, timestamps, data rates, and Windows–Linux communication.

Cal Poly Amusement Park Engineers and Designers (CAPED) — San Luis Obispo, CA

Ride Engineering Competition – Controls Team Member — *Fall 2026 – Present*

- Collaborated on the electrical and physical design of an operator control panel using pushbuttons and LED indicators to provide external ride control and system-status feedback.

FIRST Tech Challenge Robotics – Ramifications #5155 — St. Louis, MO

Programming Lead (2022–2025) & CAD/Design Lead (2023–2025) — *Aug. 2021 – May 2025*

- Led programming and CAD for a 10-member robotics team, developing 3,000+ lines of Java across 50+ source files and mentoring teammates in software, controls, and mechanical design.
- Implemented field-oriented mecanum drive, closed-loop position and heading control, and localization while integrating and debugging motors, sensors, and electromechanical subsystems.

Selected Engineering Projects

Autonomous Cryocooler Control Unit | *STM32, Altium, CAN, SPI, Simulink* — *Sep. 2026*

- Designed a preliminary spacecraft control architecture converting a 28 V spacecraft bus into 12 V cooler and 3.3 V logic rails with current/voltage monitoring and fault isolation.
- Integrated an STM32 microcontroller, CAN communication, four-wire platinum RTD sensing, and a bounded PI/PID thermal controller; modeled system response and fault conditions in Simulink.

Two-Wheel Self-Balancing Robot | *Java, IMU, Encoders, PID, LQR, Onshape* | GitHub — *Aug. 2024 – May 2025*

- Designed and fabricated a balancing prototype using a REV Control Hub, BHI260AP IMU, motor encoders, and a custom 3D-printed chassis.
- Implemented and experimentally compared PID and four-state LQR controllers using pitch angle, pitch rate, wheel position, and wheel velocity feedback.

Technical Skills

Languages: C, C++, Python, Java, MATLAB, Bash, x86-64 Assembly
Embedded & Electrical: I2C, SPI, CAN, UART, ADC, PWM, FRAM, microcontrollers, IMUs, GNSS, sensor integration
Software & Design: Git, Linux, ROS 2, NVIDIA Isaac Sim, Altium Designer, LTspice, Onshape
Controls & Lab: PID, LQR, state-space fundamentals, oscilloscope, multimeter, function generator, power supply, soldering, PCB bring-up